# Decision-Focused Learning via Tangent-Space Projection of Prediction Error

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# Overview

# Regret Gradient via Tangent-Space Projection

# Problem Setup

Given contextual $\boldsymbol{x} \in \mathbb{R}^d$, predict cost vector $\hat{\boldsymbol{c}} = f_{\boldsymbol{\theta}}(\boldsymbol{x}) \in \mathbb{R}^n$ to estimate $\boldsymbol{c}$. Given $\hat{\boldsymbol{c}}$, consider convex problem:

$$\begin{aligned} \boldsymbol{z}^*(\hat{\boldsymbol{c}}) &\in \arg\min_{\boldsymbol{z}\in\mathbb{R}^n} \{\phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}\} \\ \text{s.t.} \quad & \boldsymbol{A}_{p\times n}\boldsymbol{z} = \boldsymbol{b}, \quad \boldsymbol{G}_{m\times n}\boldsymbol{z} \leq \boldsymbol{h}. \end{aligned} \tag{1}$$

DFL defines regret under real cost vector $\boldsymbol{c}$ as:

$$R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \underbrace{[\phi(\boldsymbol{z}^*(\hat{\boldsymbol{c}})) + \boldsymbol{c}^\top \boldsymbol{z}^*(\hat{\boldsymbol{c}})]}_{\text{predicted cost}} - \underbrace{[\phi(\boldsymbol{z}^*(\boldsymbol{c})) + \boldsymbol{c}^\top \boldsymbol{z}^*(\boldsymbol{c})]}_{\text{optimal cost}}. \tag{2}$$

Training $f_{\boldsymbol{\theta}}$ requires gradient $\nabla_{\boldsymbol{\theta}} R(\hat{\boldsymbol{c}}; \boldsymbol{c})$ by

$$\frac{\mathrm{d}R(\hat{\boldsymbol{c}}; \boldsymbol{c})}{\mathrm{d}\boldsymbol{\theta}} = \frac{\partial R}{\partial \boldsymbol{z}^*} \frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} \frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}}. \tag{3}$$

And the gradient w.r.t. $\hat{\boldsymbol{c}}$ is

$$\frac{\partial R}{\partial \hat{\boldsymbol{c}}} = \left(\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}}\right)^\top (\nabla_{\boldsymbol{z}^*} \phi(\boldsymbol{z}^*) + \boldsymbol{c}). \tag{4}$$

# Active Set and Local Regularity

Main problem: active set ($\mathcal{A}(\hat{\boldsymbol{c}}) := \{i \in [m] : \boldsymbol{g}_i^\top \boldsymbol{z}^*(\hat{\boldsymbol{c}}) = h_i\}$) may change and thus $\hat{\boldsymbol{c}} \mapsto \boldsymbol{z}^*(\hat{\boldsymbol{c}})$ may be non-smooth.

## General framework (notations not consistent)

Consider $\boldsymbol{y}^*(\boldsymbol{x}) = \arg\min_{\boldsymbol{y}\in\mathcal{Y}} g(\boldsymbol{x}, \boldsymbol{y})$, $\boldsymbol{x}$ may be viewed as a parameter.

- If $\mathcal{Y} = \mathbb{R}^n$ and $g$ is smooth and convex, then optimal solution $\boldsymbol{y}^*(\boldsymbol{x})$ is given by $\nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) = 0$. Then, by implicit function theorem, $\boldsymbol{y}^*(\boldsymbol{x})$ is smooth when $\nabla^2_{\boldsymbol{yy}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x}))$ is non-singular.
- But if constrained (e.g. $\mathcal{Y} = \{\boldsymbol{y} : \boldsymbol{Gy} \leq \boldsymbol{h}\}$), then the optimality condition becoems $0 \in \nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) + \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}^*(\boldsymbol{x}))$, where $\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y})$ is the normal cone of $\mathcal{Y}$ at $\boldsymbol{y}$.
- Therefore, when $\boldsymbol{x}$ changes, the optimal solution $\boldsymbol{y}^*(\boldsymbol{x})$ may change, shifting from one face of $\mathcal{Y}$ to another, and thus the active set may change. In short, $\boldsymbol{x} \mapsto \mathcal{A}(\boldsymbol{x}) \mapsto \boldsymbol{y}^*(\boldsymbol{x})$, and $\mathcal{A}(\boldsymbol{x})$ may be discrete.
- Yet within each active set, $\boldsymbol{y}^*(\boldsymbol{x})$ is smooth, thus it is piecewise smooth.

# Appendix: Normal Cone I

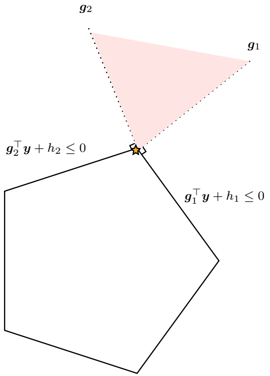
**Definition (Normal Cone).** For a closed convex set $\mathcal{Y} \subseteq \mathbb{R}^n$ and $\boldsymbol{y} \in \mathcal{Y}$, the normal cone of $\mathcal{Y}$ at $\boldsymbol{y}$ is defined as $\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}) := \{\boldsymbol{v} \in \mathbb{R}^n : \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}) \leq 0, \ \forall \boldsymbol{y}' \in \mathcal{Y}\}$.

- Tl;dr: vectors that point outward of $\mathcal{Y}$ at ref $\boldsymbol{y}$.
- For polyhedral set $\mathcal{Y} = \{\boldsymbol{y} : \boldsymbol{G}\boldsymbol{y} \leq \boldsymbol{h}\}$, the normal cone is

$$\mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}) = \left\{ \sum_{i \in \mathcal{A}(\boldsymbol{y})} \lambda_i \boldsymbol{g}_i : \lambda_i \geq 0 \right\} := \text{cone}\left\{\boldsymbol{g}_i : i \in \mathcal{A}(\boldsymbol{y})\right\}$$

  where $\mathcal{A}(\boldsymbol{y}) := \{i : \boldsymbol{g}_i^\top \boldsymbol{y} = h_i\}$ is the active set at $\boldsymbol{y}$, $\boldsymbol{g}_i^\top$ is the $i$-th row of $\boldsymbol{G}$, and cone$\{S\}$ is the conic hull of $S$.

Thus, for such constrained problem, the optimality condition is $-\nabla_{\boldsymbol{y}} g(\boldsymbol{x}, \boldsymbol{y}^*(\boldsymbol{x})) \in \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y}^*(\boldsymbol{x}))$, i.e., the descent motion can be blocked by the the combination of active constraints (*which is exactly the more general form of KKT condition*).



# Appendix: Normal Cone II

In a more convex analysis perspective, consider extended value indicator $\delta_{\mathcal{Y}}(\boldsymbol{y}) = 0$ if $\boldsymbol{y} \in \mathcal{Y}$, $+\infty$ otherwise. Then $\min_{\boldsymbol{y}\in\mathcal{Y}} g(\boldsymbol{y}) = \min_{\boldsymbol{y}\in\mathbb{R}^n}\{g(\boldsymbol{y}) + \delta_{\mathcal{Y}}(\boldsymbol{y})\} := \min_{\boldsymbol{y}\in\mathbb{R}^n} F(\boldsymbol{y})$.

**Fact:** $\partial\delta_{\mathcal{Y}}(\boldsymbol{y}) = \mathcal{N}_{\mathcal{Y}}(\boldsymbol{y})$

Proof. By subdifferential $\partial\delta_{\mathcal{Y}}(\boldsymbol{y}) = \{\boldsymbol{v} : \delta_{\mathcal{Y}}(\boldsymbol{y}') \geq \delta_{\mathcal{Y}}(\boldsymbol{y}) + \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}),\ \forall \boldsymbol{y}'\}$.

| Case | $\delta_{\mathcal{Y}}(\boldsymbol{y})$ | $\delta_{\mathcal{Y}}(\boldsymbol{y}')$ | Subgradient inequality | Consequence |
|---|---|---|---|---|
| $\boldsymbol{y} \in \mathcal{Y},\ \boldsymbol{y}' \in \mathcal{Y}$ | $0$ | $0$ | $0 \geq \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y})$ | $\boxed{\boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y}) \leq 0}$ |
| $\boldsymbol{y} \in \mathcal{Y},\ \boldsymbol{y}' \notin \mathcal{Y}$ | $0$ | $+\infty$ | $+\infty \geq \boldsymbol{v}^\top(\boldsymbol{y}' - \boldsymbol{y})$ | Automatically satisfied |
| $\boldsymbol{y} \notin \mathcal{Y},\ \boldsymbol{y}' \in \mathcal{Y}$ | $+\infty$ | $0$ | $0 \geq +\infty$ | Impossible |
| $\boldsymbol{y} \notin \mathcal{Y},\ \boldsymbol{y}' \notin \mathcal{Y}$ | $+\infty$ | $+\infty$ | $+\infty \geq +\infty$ | No additional restriction |

And the only non-trivial case is exactly the definition of normal cone.

Thus, the optimality condition can be written as $0 \in \nabla_{\boldsymbol{y}} g(\boldsymbol{y}^*) + \partial\delta_{\mathcal{Y}}(\boldsymbol{y}^*) = \partial F(\boldsymbol{y}^*)$.

# Active set and Local Regularity (cont.) I

Back to the original problem.

Given $\hat{\boldsymbol{c}}$, let $\mathcal{A}(\hat{\boldsymbol{c}}) := \{i : (\boldsymbol{G}\boldsymbol{z}^*(\hat{\boldsymbol{c}}))_i = h_i\}$ be the active set at $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$. Define the substack of the active constraint matrix

$$\boldsymbol{J}(\hat{\boldsymbol{c}}) := \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{G}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix} \in \mathbb{R}^{k\times n}, \quad k = p + |\mathcal{A}(\hat{\boldsymbol{c}})|.$$

This paper focuses on each neighbourhood, where $\mathcal{A}(\hat{\boldsymbol{c}})$ is unchanged, i.e., $\mathcal{A}(\hat{\boldsymbol{c}}) = \mathcal{A}(\hat{\boldsymbol{c}} + d\hat{\boldsymbol{c}})$ for sufficiently small perturbation $d\hat{\boldsymbol{c}}$. Thus, $\boldsymbol{J}(\hat{\boldsymbol{c}}) = \boldsymbol{J}(\hat{\boldsymbol{c}} + d\hat{\boldsymbol{c}}) := \boldsymbol{J}$ is fixed, and the solution map $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$ is smooth w.r.t. $\hat{\boldsymbol{c}}$.

**Note on $\boldsymbol{J}$**

- Since all constraints are linear, $\boldsymbol{J}(\hat{\boldsymbol{c}})$ is also the Jacobian of the active constraints at $\boldsymbol{z}^*(\hat{\boldsymbol{c}})$.
- $\boldsymbol{J} : \mathbb{R}^n \to \mathbb{R}^k$ is a linear map from decision space to active constraint space.
- $\boldsymbol{J}d\boldsymbol{z}$ can be viewed as: if $\boldsymbol{z}$ has a tiny perturbation $d\boldsymbol{z}$, then $\boldsymbol{J}d\boldsymbol{z}$ is the corresponding change of the active constraint values on each row.

# Active set and Local Regularity (cont.) II

To further ensure smothness within each region, it requires the following classical regularity conditions:

- $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*(\hat{\boldsymbol{c}}))$ is positive definite.
- LICQ (linearly independent constraint qualification): $\boldsymbol{J}(\hat{\boldsymbol{c}})$ is full row rank.
- Strict complementarity: For active inequality constraints $i \in \mathcal{A}$, the corresponding Lagrange multiplier $\mu_i^* > 0$; for inactive constraints $i \notin \mathcal{A}$, $\mu_i^* = 0$.

These assumptions ensure: solution map is locally unique, continuous, and active set is invariant under sufficiently small perturbation of $\hat{\boldsymbol{c}}$.

# Sensitivity via a Reduced KKT System I

The origin KKT system is

$$
\begin{aligned}
\nabla\phi(\boldsymbol{z}^*) + \boldsymbol{A}^\top\boldsymbol{\lambda}^* + \boldsymbol{G}^\top\boldsymbol{\mu}^* + \hat{\boldsymbol{c}} &= 0 \\
\boldsymbol{A}\boldsymbol{z}^* = \boldsymbol{b},\ \ \boldsymbol{G}\boldsymbol{z}^* \leq \boldsymbol{h}, \boldsymbol{\mu}^* &\geq 0, \\
\mathrm{diag}(\boldsymbol{\mu}^*)(\boldsymbol{G}\boldsymbol{z}^* - \boldsymbol{h}) &= 0.
\end{aligned}
\tag{5}
$$

Under the regularity conditions, focusing on local neighbourhood of $\hat{\boldsymbol{c}}$, the reduced KKT system is

$$
\nabla\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top\boldsymbol{y}^* = 0,\ \ \boldsymbol{J}\boldsymbol{z}^* = \tilde{\boldsymbol{b}}, \tag{6}
$$

where

$$
\boldsymbol{J} = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{G}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}, \quad \boldsymbol{y}^* = \begin{bmatrix} \boldsymbol{\lambda}^* \\ \boldsymbol{\mu}^*_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}, \quad \tilde{\boldsymbol{b}} = \begin{bmatrix} \boldsymbol{b} \\ \boldsymbol{h}_{\mathcal{A}(\hat{\boldsymbol{c}})} \end{bmatrix}. \tag{7}
$$

Form equation 6 into a linear system:

$$
F(\boldsymbol{z}, \boldsymbol{y}, \hat{\boldsymbol{c}}) = \begin{bmatrix} \nabla\phi(\boldsymbol{z}) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top\boldsymbol{y} \\ \boldsymbol{J}\boldsymbol{z} - \tilde{\boldsymbol{b}} \end{bmatrix} = 0. \tag{8}
$$

# Sensitivity via a Reduced KKT System II

Taking the differential of $F$ w.r.t. $(\boldsymbol{z}, \boldsymbol{y})$ and $\hat{\boldsymbol{c}}$ gives the sensitivity of the solution w.r.t. $\hat{\boldsymbol{c}}$:

$$\boxed{\begin{bmatrix} \boldsymbol{H} & \boldsymbol{J}^\top \\ \boldsymbol{J} & 0 \end{bmatrix} \begin{bmatrix} d\boldsymbol{z} \\ d\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} -d\hat{\boldsymbol{c}} \\ 0 \end{bmatrix}} \qquad \text{where } \boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*). \tag{9}$$

**Now consider its geometric interpretation.**
This equation is invertivle under the regularity conditions, and thus the first row gives $\boldsymbol{H} d\boldsymbol{z} + \boldsymbol{J}^\top d\boldsymbol{y} = -d\hat{\boldsymbol{c}}$.

$$\implies d\boldsymbol{z} = -\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y}. \tag{10}$$

Plug this in $\boldsymbol{J} d\boldsymbol{z} = 0$ gives $\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y} = -\boldsymbol{J}\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$, and thus $d\boldsymbol{y} = -(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$.
Finally, plug $d\boldsymbol{y}$ back to equation 10 gives

$$d\boldsymbol{z} = -\left(\boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top (\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1}\right) d\hat{\boldsymbol{c}} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}. \tag{11}$$

$$\boxed{\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} = -\boldsymbol{P}_H} \tag{12}$$

# Sensitivity via a Reduced KKT System III

In last page, we denote $\boldsymbol{P}_H = \boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} := \boldsymbol{\Pi}_H\boldsymbol{H}^{-1}$, where

$$\boxed{\boldsymbol{\Pi}_H := \boldsymbol{I} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}} \tag{13}$$

Mathematically, $\boldsymbol{\Pi}_H$ is a projection matrix onto the kernel of $\boldsymbol{J}$ ($\ker(\boldsymbol{J}) := \{\boldsymbol{v} : \boldsymbol{J}\boldsymbol{v} = 0\}$), under $\boldsymbol{H}$-orthogonal (i.e., by conisdering the inner product induced by $\langle \boldsymbol{u}, \boldsymbol{v}\rangle_{\boldsymbol{H}} = \boldsymbol{u}^\top\boldsymbol{H}\boldsymbol{v}$).

# Sensitivity via a Reduced KKT System IV

**Understanding of $\boldsymbol{J}$, its kernel and range**

- Kernel of $\boldsymbol{J}$: $\ker(\boldsymbol{J}) = \{\boldsymbol{v} : \boldsymbol{J}\boldsymbol{v} = 0\} \subseteq \mathbb{R}^n$, i.e., all directions that do not change the active constraint values. This is exactly the meaning of the tangent space, and denote $\mathcal{T}_{\boldsymbol{z}^*} := \ker(\boldsymbol{J})$.
- Range of $\boldsymbol{J}$: $\mathrm{range}(\boldsymbol{J}) = \{\boldsymbol{u} : \boldsymbol{u} = \boldsymbol{J}\boldsymbol{v},\ \boldsymbol{v} \in \mathbb{R}^n\} \subseteq \mathbb{R}^k$, i.e., all possible changing values of the active constraints from decision perturbations. (But we don't really care).
- Range of $\boldsymbol{J}^\top (\mathbb{R}^k \to \mathbb{R}^n)$: Recall the KKT equation 6 $\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y}^* = 0$, $\boldsymbol{y}^*$ is the Lagrange multiplier (as weights of active constraints' gradients), to balance the descent direction $-\nabla_{\boldsymbol{z}^*}\phi(\boldsymbol{z}^*) - \hat{\boldsymbol{c}}$.
  *Thus, if for arbitrary $\boldsymbol{y} \in \mathbb{R}^k$, $\boldsymbol{J}^\top \boldsymbol{y}$ is the linear combination of active constraints' gradients, i.e., the normal space of the tangent space. Denote $\mathcal{N}_{\boldsymbol{z}^*} := \mathrm{range}(\boldsymbol{J}^\top)$.*
  As a matter of fact:
  $$\mathrm{range}(\boldsymbol{J}^\top) = \ker(\boldsymbol{J})^\perp = \mathcal{T}_{\boldsymbol{z}^*}^\perp = \mathcal{N}_{\boldsymbol{z}^*}. \tag{14}$$

# Sensitivity via a Reduced KKT System V

Recall $\boldsymbol{P}_H = \boldsymbol{H}^{-1} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} := \boldsymbol{\Pi}_H \boldsymbol{H}^{-1}$.

Then revisit this $\boldsymbol{\Pi}_H$: *projection onto* $\ker(\boldsymbol{J})$ *under* $\boldsymbol{H}$*-orthogonal*. It means that, for any direction $\boldsymbol{v} \in \mathbb{R}^n$, $\boldsymbol{\Pi}_H \boldsymbol{v}$ is the closest vector to $\boldsymbol{v}$ in $\ker(\boldsymbol{J})$ under the $\boldsymbol{H}$-norm, and it will always be in the tangent space, not changing the active constraints' values.

# Sensitivity via a Reduced KKT System VI

Recall that equation 11: $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}} = -\boldsymbol{\Pi}_H \boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$ (i.e.,$\boldsymbol{P}_H : d\hat{\boldsymbol{c}} \mapsto d\boldsymbol{z}$). Then, the geometric interpretation of $\boldsymbol{P}_H$ is:

- $\boldsymbol{\Pi}_H$ is the projection of movement. Yet $d\hat{\boldsymbol{c}}$ is the perturbation of the predicted cost, not the decision.
- If this problem is unconstrained, i.e., $\min_{\boldsymbol{z}} \phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}$, then $\nabla\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} = 0$, and thus $d\boldsymbol{z} = -\nabla^2\phi(\boldsymbol{z}^*)^{-1} d\hat{\boldsymbol{c}} = -\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$. So first we know: $\boldsymbol{H}^{-1} d\hat{\boldsymbol{c}}$ maps the perturbation of predicted cost to the perturbation of decision.
- Then, given that there are active constraints, $\boldsymbol{\Pi}_H$ projects the perturbation of decision onto the tangent space, to ensure that the active constraints' values are unchanged.

# Sensitivity via a Reduced KKT System VII

Then the paper shows that,

$$\boldsymbol{J}\boldsymbol{P}_H = 0, \qquad \boldsymbol{P}_H\boldsymbol{J}^\top = 0. \tag{15}$$

- The first one is obvious, since $\boldsymbol{P}_H$ projects onto $\ker(\boldsymbol{J})$, thus $\boldsymbol{J}\boldsymbol{P}_H = 0$. *Any primal movement caused by the perturbation of predicted cost will not change the active constraints' values (in first-order approximation).*
- The second is less obvious.
  Consider a special cost perturbation $d\hat{\boldsymbol{c}} \in \mathrm{range}(\boldsymbol{J}^\top)$, i.e., $d\hat{\boldsymbol{c}} = \boldsymbol{J}^\top d\boldsymbol{y}$ for some $d\boldsymbol{y}$. Then the KKT (equation 6) gives
  $$\nabla\phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top(\boldsymbol{y}^* + d\boldsymbol{y}) = 0, \tag{16}$$
  thus the optimal solution does not change; we only need to adjust the Lagrange multiplier $\boldsymbol{y}^*$ to balance the new cost perturbation. Thus, $d\boldsymbol{z} = 0$.
  Meanwhile, equation 11 gives $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}} = -\boldsymbol{P}_H\boldsymbol{J}^\top d\boldsymbol{y} = 0$, thus $\boldsymbol{P}_H\boldsymbol{J}^\top = 0$.
  *Any cost perturbation in the normal space of the tangent space does not change the optimal solution.*

# Regret Gradient and the Projected Prediction Error I

Recall that, our key is still to solve $\frac{\partial R}{\partial \hat{\boldsymbol{c}}} = \left(\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}}\right)^\top (\nabla_{\boldsymbol{z}^*} \phi(\boldsymbol{z}^*) + \boldsymbol{c})$.

- The first term, $\frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} = -\boldsymbol{P}_H$ (equation 12) measures how much decision $\boldsymbol{z}^*$ changes w.r.t. predicted cost $\hat{\boldsymbol{c}}$. Moreover, note that $\boldsymbol{P}_H = \boldsymbol{P}_H^\top$ is symmetric.
- The second term, from reduced KKT (equation 6), $\nabla_{\boldsymbol{z}^*} \phi(\boldsymbol{z}^*) + \hat{\boldsymbol{c}} + \boldsymbol{J}^\top \boldsymbol{y}^* = 0$, thus $\nabla_{\boldsymbol{z}^*} \phi(\boldsymbol{z}^*) + \boldsymbol{c} = (\boldsymbol{c} - \hat{\boldsymbol{c}}) - \boldsymbol{J}^\top \boldsymbol{y}^*$.

Plug in the two gives

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = -\boldsymbol{P}_H(\boldsymbol{c} - \hat{\boldsymbol{c}} - \boldsymbol{J}^\top \boldsymbol{y}^*) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}) + \underbrace{\boldsymbol{P}_H \boldsymbol{J}^\top}_{=0} \boldsymbol{y}^* = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}). \tag{17}$$

Denote $\boldsymbol{e} = \hat{\boldsymbol{c}} - \boldsymbol{c}$ as the prediction error, then the key formula, i.e. projected prediction error, is

$$\boxed{\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H \boldsymbol{e}} \tag{18}$$

Compared with MSE gradient $\nabla_{\hat{\boldsymbol{c}}} \mathcal{L}_{\text{MSE}} = \hat{\boldsymbol{c}} - \boldsymbol{c} = \boldsymbol{e}$, it implies: *each direction of pred.error needs to be adjusted.*

# Regret Gradient and the Projected Prediction Error II

## Corollary (Normal Filtering and Normal Invariance)

*Let $\mathcal{N} = \mathrm{range}(\boldsymbol{J}^\top)$ be the normal space of the tangent space $\mathcal{T}$. For any $\boldsymbol{\delta}_N \in \mathcal{N}$,*

$$\boldsymbol{P}_H \boldsymbol{\delta}_N = 0. \tag{19}$$

*If active set is unchanged, then for all $\alpha \in \mathbb{R}$,*

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}} + \alpha \boldsymbol{\delta}_N; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} + \alpha \boldsymbol{\delta}_N - \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}) + \alpha \underbrace{\boldsymbol{P}_H \boldsymbol{\delta}_N}_{=0} = \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}). \tag{20}$$

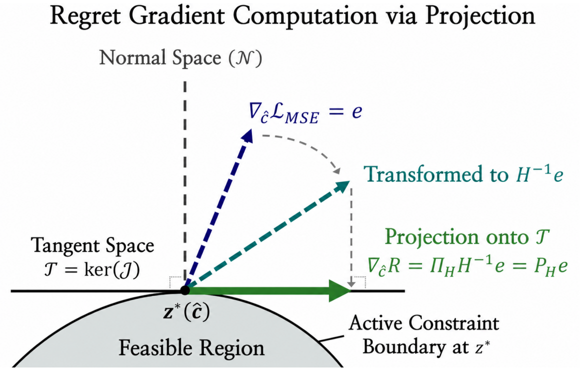
- Here, equation 19 is basically the same as $\boldsymbol{P}_H \boldsymbol{J}^\top = 0$.
- Then, the corollary shows that, since $\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c})$, then if there is anything normal in the prediction error: $\boldsymbol{e} = \boldsymbol{e}_T + \boldsymbol{\delta}_N$, then all those normal components $\boldsymbol{\delta}_N$ will be filtered out, contributing nothing to the regret gradient.

**Thus, the regret gradient is invariant to any normal perturbation of the predicted cost $\hat{c}$, as long as the active set is unchanged.**

# Regret Gradient and the Projected Prediction Error III

Finally, the key takeaway is this decomposition of the regret gradient:

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H \boldsymbol{e} = \Pi_H \boldsymbol{H}^{-1} \boldsymbol{e} = \underbrace{\left(\boldsymbol{I} - \boldsymbol{H}^{-1} \boldsymbol{J}^\top (\boldsymbol{J} \boldsymbol{H}^{-1} \boldsymbol{J}^\top)^{-1} \boldsymbol{J}\right)}_{\text{projection onto tangent space}} \underbrace{\boldsymbol{H}^{-1} \boldsymbol{e}}_{\text{preconditioning by Hessian}} . \tag{21}$$



# Computing the Projected-Error Gradient

# General Framework of PEAR

From last section,

$$\nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}).$$

The general update is still the classic first-order gradient descent:

$$\begin{aligned} \frac{d R}{d \boldsymbol{\theta}} &= \frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}}^{\top} \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) \\ &= \left( \frac{\partial \hat{\boldsymbol{c}}}{\partial \boldsymbol{\theta}} \right)^{\top} \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c}). \end{aligned} \tag{22}$$

The key is to compute

$$\boldsymbol{g} := \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c}) = \boldsymbol{P}_H(\hat{\boldsymbol{c}} - \boldsymbol{c})$$

efficiently, rather than forming $\boldsymbol{P}_H \in \mathbb{R}^{n \times n}$ explicitly.

**Algorithm 1 PEAR: Projected Error as Regret-gradient**

**Input:** Predicted costs $\hat{\boldsymbol{c}} = f_{\boldsymbol{\theta}}(\boldsymbol{x})$, true costs $\boldsymbol{c}$, problem data $(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{G}, \boldsymbol{h})$
**Input:** Hessian $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*)$ (or $\boldsymbol{H} = \lambda \boldsymbol{I}$ for LP), injection strength $\beta \in [0, 1]$
**Output:** Gradient signal $\boldsymbol{g} = \nabla_{\hat{\boldsymbol{c}}} R(\hat{\boldsymbol{c}}; \boldsymbol{c})$

// **Forward: Active-Set Detection**
Solve (1) with cost $\hat{\boldsymbol{c}}$ to obtain primal-dual pair $(\boldsymbol{z}^*, \boldsymbol{y}^*)$
Identify active set $\mathcal{A}$ via complementary slackness (16)
Form active constraint Jacobian $\boldsymbol{J} \leftarrow \left[ \boldsymbol{A}^{\top}, \boldsymbol{G}_{\mathcal{A}}^{\top} \right]^{\top}$
// **Backward: Compute $\boldsymbol{P}_H \boldsymbol{e}$ via Reduced System**
Compute prediction error $\boldsymbol{e} \leftarrow \hat{\boldsymbol{c}} - \boldsymbol{c}$
Compute $\boldsymbol{x} \leftarrow \boldsymbol{H}^{-1} \boldsymbol{e}$ and $\boldsymbol{r} \leftarrow \boldsymbol{J} \boldsymbol{x}$
Solve $(\boldsymbol{J} \boldsymbol{H}^{-1} \boldsymbol{J}^{\top}) \boldsymbol{v} = \boldsymbol{r}$ for $\boldsymbol{v}$
Compute projected gradient $\boldsymbol{g} \leftarrow \boldsymbol{x} - \boldsymbol{H}^{-1} \boldsymbol{J}^{\top} \boldsymbol{v}$
// **Optional: Normal injection for LP**
**if** $\beta > 0$ **then**
Compute normal component $\boldsymbol{n} \leftarrow \boldsymbol{H}^{-1} \boldsymbol{J}^{\top} \boldsymbol{v}$
$\boldsymbol{g} \leftarrow \boldsymbol{g} + \beta \cdot \frac{\|\boldsymbol{g}\|}{\|\boldsymbol{n}\|} \cdot \boldsymbol{n}$
**end if**
**return** $\boldsymbol{g}$

# Forward Pass: Active-Set Detection

Not too much theory.

1. First, use a solver `OSQP` to solve the optimization problem $\boldsymbol{z}^*(\hat{\boldsymbol{c}}) \in \arg\min_{\boldsymbol{z}} \phi(\boldsymbol{z}) + \hat{\boldsymbol{c}}^\top \boldsymbol{z}$ s.t. $\boldsymbol{Az} = \boldsymbol{b}, \boldsymbol{Gz} \leq \boldsymbol{h}$, and obtain the primal-dual pair $(\boldsymbol{z}^*, \boldsymbol{y}^*)$.
2. Then, identify the active set $\mathcal{A}(\hat{\boldsymbol{c}})$ via complementary slackness: $\mathcal{A}(\hat{\boldsymbol{c}}) = \{i : (\boldsymbol{Gz}^*(\hat{\boldsymbol{c}}))_i = h_i\}$.
3. Finally, form the active constraint Jacobian $\boldsymbol{J} = [\boldsymbol{A}^\top, \boldsymbol{G}_{\mathcal{A}}^\top]^\top$.

# Backward Pass: Efficient Gradient Computation I

Recall, given a perterbation in predicted cost $d\hat{\boldsymbol{c}}$, the corresponding perturbation in optimal solution is $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}$. Now, specify the perturbation value as the prediction error

$$d\hat{\boldsymbol{c}} \leftarrow -\boldsymbol{e}. \tag{23}$$

Plug in this specific perturbation into the sensitivity KKT system (equation 9) gives:

$$\begin{bmatrix} \boldsymbol{H} & \boldsymbol{J}^\top \\ \boldsymbol{J} & 0 \end{bmatrix} \begin{bmatrix} d\boldsymbol{z} \\ d\boldsymbol{y} \end{bmatrix} = \begin{bmatrix} \boldsymbol{e} \\ 0 \end{bmatrix} \tag{24}$$

Computing equation 24 gives:

$$d\boldsymbol{z} = \underbrace{\boldsymbol{H}^{-1}\boldsymbol{e}}_{\text{free movement}} - \underbrace{\boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y}}_{\text{constraint correction}}, \qquad \underbrace{\boldsymbol{J}d\boldsymbol{z}}_{\text{active constraint}} = 0. \tag{25}$$

Solving this linear system gives the projected prediction error $d\boldsymbol{z} = \boldsymbol{P}_H\boldsymbol{e}$.

# Backward Pass: Efficient Gradient Computation II

It solves with some tricks as follows. Combining the two gives the reduced system:

$$\boxed{\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top d\boldsymbol{y} = \boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{e}} \tag{26}$$

To simplify notation, $\boldsymbol{x} := \boldsymbol{H}^{-1}\boldsymbol{e} \in \mathbb{R}^n$, $\boldsymbol{r} := \boldsymbol{J}\boldsymbol{x} \in \mathbb{R}^k$, and $\boldsymbol{v} := d\boldsymbol{y} \in \mathbb{R}^k$. Then the reduced system (only involves $k$ variables) is

$$\boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top \boldsymbol{v} = \boldsymbol{r} \tag{27}$$

Then once $\boldsymbol{v} = d\boldsymbol{y}$ is solved, recall that $\boldsymbol{g} = \boldsymbol{P}_H\boldsymbol{e}$ (equation 18), $d\boldsymbol{z} = -\boldsymbol{P}_H d\hat{\boldsymbol{c}}$ (equation 11), and $d\hat{\boldsymbol{c}} = -\boldsymbol{e}$ (equation 23). These three equations give $\boldsymbol{g} = d\boldsymbol{z}$. Further by $d\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top\boldsymbol{v}$ (equation 25), it gives

$$\boxed{\boldsymbol{g} = d\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top\boldsymbol{v}} \tag{28}$$

# LP: Quadratic Smoothing and Normal Injection I

LP is problematic. Consider

$$\boldsymbol{z}^* \in \arg\min_{\boldsymbol{z}} \hat{\boldsymbol{c}}^\top \boldsymbol{z}, \quad \text{s.t. } \boldsymbol{A}\boldsymbol{z} = \boldsymbol{b}, \boldsymbol{G}\boldsymbol{z} \leq \boldsymbol{h}. \tag{29}$$

- Recall the previous analysis is based on $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*)$. However, for LP, $\boldsymbol{H} = 0$.
- Since LP's optimum usually lies on some vertex, thus the solution map $\hat{\boldsymbol{c}} \mapsto \boldsymbol{z}^*(\hat{\boldsymbol{c}})$ is piecewise constant, and the regret gradient $\frac{\partial R}{\partial \hat{\boldsymbol{c}}} = \frac{\partial R}{\partial \boldsymbol{z}^*} \frac{\partial \boldsymbol{z}^*}{\partial \hat{\boldsymbol{c}}} = 0$ almost everywhere.

Thus, here use quadratic smoothing:

$$\boldsymbol{z}^* \in \arg\min_{\boldsymbol{z}} \frac{\lambda}{2}\|\boldsymbol{z}\|_2^2 + \hat{\boldsymbol{c}}^\top \boldsymbol{z}, \quad \text{s.t. } \boldsymbol{A}\boldsymbol{z} = \boldsymbol{b}, \boldsymbol{G}\boldsymbol{z} \leq \boldsymbol{h}. \tag{30}$$

Then $\phi(\boldsymbol{z}) = \frac{\lambda}{2}\|\boldsymbol{z}\|_2^2$, and $\boldsymbol{H} = \nabla^2 \phi(\boldsymbol{z}^*) = \lambda \boldsymbol{I}$ is positive definite.

# LP: Quadratic Smoothing and Normal Injection II

Based on this smoothing, the previous algorithm can be applied (recall $\boldsymbol{v} = d\boldsymbol{y}$):

$$\text{(\textit{equation} 26)} \quad \boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{J}^\top \boldsymbol{v} = \boldsymbol{J}\boldsymbol{H}^{-1}\boldsymbol{e} \qquad \Longrightarrow \boldsymbol{J}\boldsymbol{J}^\top \boldsymbol{v} = \boldsymbol{J}\boldsymbol{e} \tag{31}$$

$$\text{(\textit{equation} 28)} \quad \boldsymbol{g} = d\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{H}^{-1}\boldsymbol{J}^\top \boldsymbol{v} \qquad \Longrightarrow \boldsymbol{g} = \frac{1}{\lambda}(\boldsymbol{e} - \boldsymbol{J}^\top \boldsymbol{v}) \tag{32}$$

# LP: Quadratic Smoothing and Normal Injection III

## Interpretation of $\boldsymbol{J}\boldsymbol{J}^\top \boldsymbol{v} = \boldsymbol{J}\boldsymbol{e}$

- When $\boldsymbol{J}$ is full row rank, equation 31 gives $\boldsymbol{v} = (\boldsymbol{J}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{e}$, then $\underline{\boldsymbol{J}^\top \boldsymbol{v}} = \boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{e} := \boldsymbol{\Pi}_{\mathcal{N}}\boldsymbol{e} := \underline{\boldsymbol{e}_{\mathcal{N}}}$, where $\boldsymbol{\Pi}_{\mathcal{N}} = \boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}$ is the projection onto the normal space $\mathcal{N} = \mathrm{range}(\boldsymbol{J}^\top)$, and $\boldsymbol{e}_{\mathcal{N}}$ is the normal component of $\boldsymbol{e}$.
- Accordingly, $\boldsymbol{\Pi}_{\mathcal{T}} = \boldsymbol{I} - \boldsymbol{\Pi}_{\mathcal{N}}$ is the projection onto the tangent space $\mathcal{T} = \ker(\boldsymbol{J})$. This can also be seen from equation 31 that $\boldsymbol{J}\boldsymbol{e} - \boldsymbol{J}\boldsymbol{J}^\top \boldsymbol{v} = \boldsymbol{J}(\boldsymbol{e} - \boldsymbol{J}^\top \boldsymbol{v}) = 0$, thus $\underline{\boldsymbol{e} - \boldsymbol{J}^\top \boldsymbol{v}} = \boldsymbol{e} - \boldsymbol{J}^\top(\boldsymbol{J}\boldsymbol{J}^\top)^{-1}\boldsymbol{J}\boldsymbol{e} = \boldsymbol{\Pi}_{\mathcal{T}}\boldsymbol{e} := \underline{\boldsymbol{e}_{\mathcal{T}}} \in \ker(\boldsymbol{J})$.
- Therefore, $\boldsymbol{e} \equiv \boldsymbol{\Pi}_{\mathcal{T}}\boldsymbol{e} + \boldsymbol{\Pi}_{\mathcal{N}}\boldsymbol{e} := \boldsymbol{e}_{\mathcal{T}} + \boldsymbol{e}_{\mathcal{N}}$. And thus equation 32 equivalents to $\boldsymbol{g} = \frac{1}{\lambda}(\boldsymbol{e} - \boldsymbol{e}_{\mathcal{N}}) = \frac{1}{\lambda}\boldsymbol{e}_{\mathcal{T}}$, which gives:

$$\underbrace{\boldsymbol{g}}_{\text{tangent regret grad}} = \underbrace{\lambda^{-1}\boldsymbol{e}}_{\text{scaled MSE grad}} - \underbrace{\lambda^{-1}\boldsymbol{e}_{\mathcal{N}}}_{\text{scaled normal component}} \tag{33}$$

# LP: Quadratic Smoothing and Normal Injection IV

However, even after smoothing, the regret gradient $\boldsymbol{g}$ may still be weak and nearly constant over large regions.

## One possible understanding:

- $\dim \mathcal{T} = \dim \ker(\boldsymbol{J}) = n - k$ (given $\boldsymbol{J} \in \mathbb{R}^{k \times n}$ full row rank), and $\dim \mathcal{N} = \dim \operatorname{range}(\boldsymbol{J}^\top) = k$. Recall that $k$ is the number of independent active constraints, and for LP's where usually optimum lies on some vertex, $k$ is usually large, and thus $\dim \mathcal{T} \ll \dim \mathcal{N}$.
- Then, heuristically, $\mathbb{E}(\|\boldsymbol{e}_{\mathcal{T}}\|^2) \propto \dim \mathcal{T} \ll \dim \mathcal{N} \propto \mathbb{E}(\|\boldsymbol{e}_{\mathcal{N}}\|^2)$, thus $\|\boldsymbol{e}_{\mathcal{T}}\| \ll \|\boldsymbol{e}_{\mathcal{N}}\|$, and thus $\boldsymbol{g} = \frac{1}{\lambda}\boldsymbol{e}_{\mathcal{T}}$ is weak.

# LP: Quadratic Smoothing and Normal Injection V

To mitigate this, consider

$$\boldsymbol{g}_{\text{inj}} := \boldsymbol{g} + \beta \cdot \frac{\|\boldsymbol{g}\|}{\|\lambda^{-1}\boldsymbol{e}_{\mathcal{N}}\|} \cdot \lambda^{-1}\boldsymbol{e}_{\mathcal{N}} = \frac{1}{\lambda}\boldsymbol{e}_{\mathcal{T}} + \beta \cdot \frac{\|\boldsymbol{g}\|}{\|\boldsymbol{e}_{\mathcal{N}}\|} \cdot \boldsymbol{e}_{\mathcal{N}}. \tag{34}$$

## Another viewpoint:

- From equation 16, any normal perturbation of the predicted cost $\hat{\boldsymbol{c}}$ will not change the optimal solution, and will be absorbed by the Lagrange multiplier $\boldsymbol{y}^*$.
- Yet according to the complementary slackness, a constraint is active iff $\boldsymbol{y}_i^* > 0$, or equivalently, $\boldsymbol{y}_i^* = 0$ iff the constraint is inactive.
- Thus, a change in the normal direction will change the Lagrange multiplier $\boldsymbol{y}^*$, and thus the active set may change.
- This gives LP a chance to escape from the current active set, and thus escape from the current vertex, and explore other vertices.

# Experiments

# Synthetic LP Benchmarks

First, the paper considers two standard synthetic LP benchmarks.

1. **Shortest Path.** On a directed $5 \times 5$ grid, the model predicts 40 edge costs, and the downstream LP selects the minimum-cost path.
2. **Knapsack.** The model predicts the values of 100 items. PEAR and LAVA use the LP relaxation during training, while all methods are evaluated on the original 0–1 problem.

The feature vector is $\boldsymbol{x} \sim \mathcal{N}(0, \boldsymbol{I}_5)$, and the true costs are generated by a polynomial mapping with degree $\deg \in \{2, 4, 6, 8\}$. Since the predictor is linear, a larger degree means a more nonlinear and harder prediction problem.

- Each task contains 1,000 training, 500 validation, and 500 test samples. Results are averaged over five random seeds.
- Baselines include two-stage MSE and four DFL methods: SPO$+$, PFYL, DBB, and LAVA.
- The metrics are normalized regret and training time. Additional tests inject multiplicative noise $\epsilon \in \{0.1, 0.3, 0.5\}$ into the cost mapping.

Thus, this experiment mainly tests whether PEAR can retain decision quality on LPs while computing the backward signal efficiently.

Source: Lee et al. (2026), Sec. 5.1 and Apps. C.1–C.2, E.1.

# Synthetic LP Benchmarks

The following table keeps several representative settings from the full results.

| Task | Deg. | Regret (%, ↓) | | Time (s, ↓) | |
|---|---|---|---|---|---|
| | | PEAR | Best competitor | PEAR | Competitor |
| Shortest Path | 4 | 0.774 | **0.761** (SPO+) | **23.5** | 41.8 |
| Shortest Path | 8 | **4.246** | 4.403 (SPO+) | **38.4** | 43.2 |
| Knapsack | 4 | **0.315** | 0.381 (SPO+) | **305.2** | 600.0 |
| Knapsack | 8 | **0.437** | 0.763 (SPO+) | **271.7** | 600.0 |

- On Shortest Path, PEAR is best or tied at degrees 2, 6, and 8. At degree 4, SPO+ is slightly better in regret (0.761 versus 0.774), but PEAR is about 1.8× faster.
- On Knapsack, PEAR gives the lowest regret at every degree. At degree 8, its regret is 0.437, compared with 0.763 for SPO+; SPO+, PFYL, and DBB all reach the 600-second limit.
- Under multiplicative noise, PEAR remains best on both tasks. Even at $\epsilon = 0.5$, it obtains regret 10.38 on Shortest Path and 1.58 on Knapsack.

In short, PEAR is best or tied in seven of the eight clean task–degree settings, with the clearest advantage on Knapsack and at larger polynomial degrees.

Source: Lee et al. (2026), Tables 1–2.

# Real-World QP: Mean–Variance Optimization

Next, the paper considers a real-world QP based on S&P 100 constituents from 2010–2024. The downstream problem is

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \frac{\lambda}{2} \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} - \hat{\boldsymbol{\mu}}^\top \boldsymbol{w} \\ \text{s.t.} \quad & \mathbf{1}^\top \boldsymbol{w} = 1, \qquad \boldsymbol{w} \geq 0, \end{aligned} \tag{35}$$

where $\lambda = 2.0$, so the optimizer constructs a fully invested, long-only mean–variance portfolio.

- DLinear with RevIN uses a 63-day return window to predict the next 21 days. The data are split chronologically into 70% / 10% / 20%.
- The portfolio is rebalanced every 21 trading days, with a 0.1% transaction cost applied to turnover.
- Baselines are two-stage MSE, QPTH, and CvxpyLayers. Evaluation includes regret, training time, returns, Sharpe ratio, and maximum drawdown.

For PEAR, OSQP is used in the forward pass to identify the active set, while the reduced-system backward pass is implemented through batched GPU linear algebra. This directly tests the computational claim from the previous section on a QP with $\boldsymbol{H} = \lambda \boldsymbol{\Sigma}$.

Source: Lee et al. (2026), Sec. 5.2 and Apps. C.3, E.2.

# Real-World QP: Mean–Variance Optimization

Table 3 reports both optimization metrics and realized portfolio performance.

| Method | Regret ↓ (%) | Time ↓ (s) | Cum. Ret. ↑ (%) | Ann. Ret. ↑ (%) | Sharpe ↑ | MDD ↑ (%) |
|---|---|---|---|---|---|---|
| MSE | 90.80 | **33.0** | 89.22 | 37.51 | 0.92 | -40.00 |
| QPTH | 87.85 | 147.6 | 139.77 | 47.88 | 1.15 | -34.94 |
| CvxpyLayers | 87.36 | 321.9 | 134.88 | 47.39 | 1.15 | -36.47 |
| **PEAR** | **85.38** | **122.3** | **184.19** | **58.84** | **1.44** | **-29.35** |

Means over five seeds; standard deviations are omitted for readability. For MDD, closer to zero is better.

- PEAR gives the lowest regret, 85.38%, and is faster than both differentiable QP layers: 122.3 seconds versus 147.6 for QPTH and 321.9 for CvxpyLayers.
- It also gives the best realized portfolio metrics: cumulative return 184.19%, annualized return 58.84%, Sharpe ratio 1.44, and maximum drawdown $-29.35\%$.
- MSE is the fastest because it does not differentiate through the downstream problem, but its decision quality and portfolio performance are worse.

Thus, in this experiment, lower decision-focused regret is accompanied by both higher realized return and better downside-risk control.

Source: Lee et al. (2026), Table 3.

# Robustness Under Constraint Shifts

Finally, the paper asks whether the learned predictor still works when the downstream constraints change after training. The cost or return process is fixed, and only the feasible region is modified.

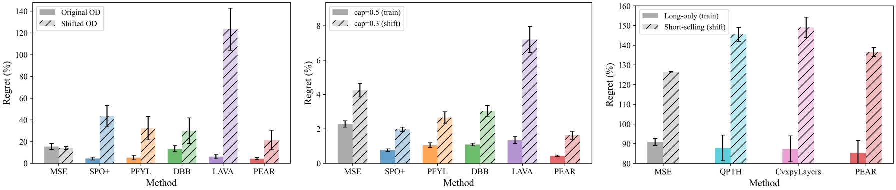
| Task | Training constraint | Test-time constraint |
|---|---|---|
| Shortest Path | Directed corner-to-corner route | New origin–destination pair and edge direction |
| Knapsack | Capacity ratio $\rho = 0.5$ | $\rho \in \{0.3, 0.5, 0.7, 0.9\}$ |
| Portfolio QP | Long-only ($\ell = 0$) | $\ell \in \{-0.1, -0.3, -0.5, -1.0\}$ |

- Shortest Path changes the routing direction and origin–destination pair on the same grid, testing whether the model learns edge costs rather than memorizing training paths.
- Knapsack changes the capacity ratio, making the feasible region tighter or looser. The main-text figure uses the tighter shift $0.5 \to 0.3$.
- Portfolio QP permits short selling through negative lower bounds. Unlike the first two representative shifts, this expands the feasible region.

For the LP tasks, the main comparison uses degree 8, i.e., the hardest polynomial setting. The metric is normalized regret under the shifted constraint.

Source: Lee et al. (2026), Sec. 5.3 and Apps. D, F.

# Robustness Under Constraint Shifts




*(a)* Origin-Destination shift (Shortest Path) *(b)* Capacity shift (Knapsack) *(c)* Short-selling shift (MVO)

- **Shortest Path.** MSE is best overall. Among DFL methods, PEAR degrades least, while SPO+, PFYL, DBB, and especially LAVA deteriorate substantially. This suggests that some DFL methods learn paths specialized to the training direction.
- **Knapsack.** PEAR gives the lowest regret when capacity tightens. The appendix shows the same pattern across the wider capacity sweep and all polynomial degrees.
- **Portfolio QP.** MSE is again best, while PEAR is the best DFL method. The DFL gap is smaller here, likely because the QP methods use closely related KKT information.

In short, PEAR is consistently the most robust method within the DFL family. However, MSE remains better on the Shortest Path and Portfolio shifts, so in-distribution decision quality and generalization to a new feasible region should be viewed separately.

Source: Lee et al. (2026), Fig. 2 and Apps. F.1–F.3.

# References

Lee, J., Jin, S., and Lee, Y. (2026). Decision-focused learning via tangent-space projection of prediction error.